Chapter
12

Microbial Genetics

Learning Objectives

After studying this chapter the students will be able to,

- *Define gene, genome, genetic code, genotype, phenotype, mutagen, wildtype*
- *Describe transcription and translation*
- *Classify mutations and its types and Understand how mutants are formed*
- *Know the mode of action of physical and chemical mutagens*
- *Identify the purpose of and outline the procedure for Ames test*
- *Compare the gene transfer mechanisms*
- *Know the types of cloning vectors used in genetic engineering*
- *Describe how plasmids and bacteriophages are used to transfer foreign DNA*
- *Explain the role of restriction enzymes in recombinant DNA technology*
- *Know the types of restriction enzymes*
- *Understand agarose gel electrophoresis and PCR techniques*
- *Explain RAPD and RFLP*

Chapter Outline

12.1 Concept of Gene

12.2 Transcription

12.3 Genetic Code

12.4 Translation

12.5 Types of Mutation

12.6 Formation of Mutants

12.7 Transfer of Genetic Material

12.8 Recombinant DNA Technology

12.9 Vectors – Types and Characteristics

12.10 Restriction Enzymes

12.11 Techniques in Genetic Engineering

12.1 Concept of Gene

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The fundamental unit of information in living systems is the **gene**. **Genome** is the set of all genes and genetic signals of a cell. The information contained in genes is converted to molecules that determine the metabolism, structure and form of microorganisms. Gene is expressed through a sequence of events. A gene can be defined biochemically as a segment of DNA (or, in a few cases, RNA) that encodes the information required to produce a functional biological product. The final product is usually a protein. Not all genes are involved in protein synthesis; some code instead for rRNA and tRNA.

The central dogma of molecular biology, comprises the three major processes (Figure 12.1). The first is **replication**, the

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